
CLINTON ENWEREM

enwerem@umd.edu | 301-416-9430 | clintonenwerem.com | GitHub: @coenwerem | LinkedIn | Scholar

SUMMARY

Robotics Research & Software Engineer, Ph.D. (ECE, University of Maryland, 2026). Specialties in dexterous manipulation, risk-aware and safety-critical control, generative grasp models, reinforcement and imitation learning, and real-time QP control. **5 years** building manipulation software for real multifingered hardware. 7 peer-reviewed and accepted papers. Seeking research scientist, research engineer, or applied ML roles.

SKILLS

Robotics: ROS/ROS2, MuJoCo, Drake, Isaac Sim, Isaac Lab, Pinocchio, MoveIt2; **Control & Safe Learning:** Control Barrier Functions, Safety Filters, Safe & Constrained Reinforcement Learning, Model Predictive Control (MPC), QP-based Control, Belief-Space Planning, Sim-to-Real Transfer; **Robot Learning & Generative Models:** Imitation Learning & Behavior Cloning, Diffusion Policies (DP3), Flow-Matching Generative Models, Reinforcement Learning (PPO); **ML / Perception:** PyTorch, TensorFlow, OpenCV; **Programming:** Python, C++, MATLAB, Bash; **Optimization:** CasADi, CVXPY, qpsolvers, IPOPT, Gurobi; **Dev Tools:** git, Docker; **Robots & EoATs:** Fairino FR3, xArm7, Unitree G1 (simulation), LEAP Hand v1, RealHand L6.

WORK EXPERIENCE

Graduate Research Assistant

Institute for Systems Research (ISR), University of Maryland

Aug. 2021 – Present
College Park, MD

- Build and maintain the lab's Python and C++ grasp planning and closed-loop control stack. Deploy the stack on Fairino FR3 and xArm7 arms with LEAP Hand v1 and RealHand L6 end-effectors.
- Build expert data collection pipelines over synchronized tactile, proprioceptive, and visual signals. Train behavior cloning policies, MLP, LSTM, and DP3 (3D Diffusion Policy), for grasp stabilization under partial tactile observability.

Robotics Trainee

Robotics & Artificial Intelligence Nigeria

Jan. 2020 – Feb. 2021
Ibadan, Nigeria

- Built ROS-based visual-SLAM, state-estimation, and flight-control software for a low-cost UAV on embedded hardware.

PROJECTS

GDF, Planner-Free Certified Grasping (Project Page) | Python, MuJoCo, CasADi

2026

- Built a planner-free grasp execution controller from a softmin distance field over certified pregrasp configurations, filtered by a CBF-CLF quadratic program. Proved forward invariance of the safe set. Lifted **46 of 50** objects, retaining a median **94%** of the synthesized quality margin.
- Ported the controller to a **Unitree G1 humanoid** and a fixed-base arm by swapping only the robot description. QP solves in **0.09 ms** against a 20 ms control step.

EquiDexFlow, SE(3)-Equivariant Grasp Generation (Project Page, Code) | Python, PyTorch

Jun. 2026

- Trained an SE(3)-equivariant flow-matching generative model for grasp synthesis on 8,100 certified grasps across 81 objects, with friction-cone compliance holding by construction. **Zero** friction violations. Retargeted to a physical LEAP Hand by IK alone.

VNB-Grasp, Variational Neural Belief Parameterizations (Code) | Python, PyTorch, MuJoCo

Apr. 2026

- Designed a differentiable Gaussian-mixture contact belief with pathwise CVaR gradients, and released a MuJoCo benchmark scoring friction sensitivity and perturbation tolerance. Cut belief propagation **over 10x**.

URDF Kinematics Toolkit (Code) | Python

Oct. 2025

- Built a URDF parsing and kinematics library with forward and inverse kinematics, damped least-squares Jacobian resolution, condition-number singularity margins, and kinematics-aware configuration-space sampling.

SELECTED PUBLICATIONS

- **C. Enwerem**, J. S. Baras, C. Belta. "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Manuscript*, 2026. Project page.
- **C. Enwerem**, J. S. Baras, C. Belta. "FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis." *Preprint*, 2026. [arXiv:2607.25049](https://arxiv.org/abs/2607.25049).
- Also to appear 2026 at IROS and CDC, plus 5 peer-reviewed first-author papers (CoDIT'23, L-CSS'24, ECC'24, CDC'24, CDC'25).

EDUCATION

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present

B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0.

Aug. 2018